Slice generator: the previous shipped scheme, the three two-stage variants, the vertex-residual variant and the combined variant, adopted (2026-11-09)

Each row generates one slice record from the truth; the PFP generator (unchanged, run m4_official_m3w0) then builds the PFP set from it. Numbers: CC1p0pi acceptance on test jobs 0-199.

1. Previous shipped slice generator (run m2_official_m3, shipped until 2026-11-09)

Truth input
80 summary features
+ 64-d particle-encoder z

(no Geant4 quantities)

→ **Topology classifier**
softmax MLP, 121 classes of
(n_pfp, n_prong, n_trk,
n_proton, n_shower)
sampled first, from truth
-> the PFP counts

→ **Presence classifiers**
GBDT on truth + class:
has 2nd track
has two-track block
has charge spheres 1-4 cm
+ point masses

→ **One flow: all slice values**
~50 numbers: vertex, scores,
longest track, 2nd track,
two-track block, spheres
masked to the groups present
conditioned on truth + class

→ **Result (jobs 0-199): purity 0.394, eff 0.191**
(real 0.435 / 0.198).
Counts, presence and the longest track are drawn
in separate pieces, each from the truth alone.
Lost: exiting longest track <-> fewer PFPs.
CC Npi at n_pfp = 3, longest end in FV: 0.49
(real 0.44) -> too many CC Npi pass 2prong-contained.

2. Variant A tried (run m2s): two stages, counts as continuous numbers in stage 1

Truth input
80 summary features
+ 64-d particle-encoder z

(no Geant4 quantities)

→ **Stage-1 flow (one joint sample)**
always-present values: vertex,
scores, longest track (+ wall flag)
+ the 5 counts as continuous
numbers, rounded afterwards
+ 6 presence flags

→ **Hard rules**
counts nested
n_trk >= 2 -> 2nd track
two-track block only
for n_prong = n_trk = 2
fixes: < 1% of slices

→ **Stage-2 flow: optional groups**
2nd track, two-track block,
charge spheres
conditioned on truth + stage-1
output; masked by the
sampled presence

→ **Result: the link closes (0.442 vs 0.439),**
pion range momentum on the curve 93% (real 93%).
But rounding 5 continuous numbers does not keep
the real count combinations: 17% of slices get a
tuple real slices never have; (3,2,2,*,0) is 0.34
of n_pfp = 3 (real 0.44). The PFP generator then
builds fewer 2-track sets. Purity 0.396, eff 0.165.

3. Variant B tried (run m2o): two stages, counts as a one-hot class in stage 1

Truth input
80 summary features
+ 64-d particle-encoder z

(no Geant4 quantities)

→ **Stage-1 flow (one joint sample)**
always-present values: vertex,
scores, longest track (+ wall flag)
+ the counts as a one-hot over the
121 classes: noisy in training,
argmax at sampling
+ 6 presence flags

→ **Hard rules**
counts always a real tuple
n_trk >= 2 -> 2nd track
two-track block only
for n_prong = n_trk = 2
fixes: 3.3% / 1.1%

→ **Stage-2 flow: optional groups**
as in row 2

→ **Result: every tuple real; (3,2,2,*,0) 0.435**
(real 0.439); P(PFP 2prong | n_pfp = 3) 0.966 (0.971).
But the argmax over 121 entries overweights high
multiplicity (n_pfp >= 6: 0.115 vs 0.084), the
longest-track length is generated less well, and
the link is lost again (0.515).
Purity 0.371, eff 0.170.

4. Variant C tried (run m2c): two stages, counts from a classifier that reads the generated stage 1

Truth input
80 summary features
+ 64-d particle-encoder z

(no Geant4 quantities)

→ **Stage-1 flow (one joint sample)**
always-present values: vertex,
scores, longest track (+ wall)
+ 6 presence flags
no counts

→ **Topology classifier**
softmax over the 121 classes
input: truth + generated stage-1
values and presence flags
classes incompatible with the
sampled presence set to 0
sampled -> the PFP counts

→ **Stage-2 flow: optional groups**
as in row 2; conditioned on
truth + stage-1 output + class

→ **Result: counts right, link still missing**
Counts fixed: tuple distance to real 0.017 (shipped 0.015),
(3,2,2,*,0) 0.436 (real 0.439), P(PFP 2prong | n_pfp = 3)
0.970 (0.971), no slice without a compatible class.
But the link is not reproduced: CC Npi at n_pfp = 3,
longest end in FV 0.488 (real 0.439, shipped 0.49).
Purity 0.396, eff 0.191 = the shipped chain.
At the M2 level no variant has the link (0.48-0.52 vs
0.446); M2s's chain 0.442 came from its distorted counts.

5. Variant D tried (run m2v): the shipped scheme with the vertex generated as the residual to the true vertex

Truth input
80 summary features
+ 64-d particle-encoder z
+ the true vertex, added
back after the flow
(no Geant4 quantities)

→ **Topology classifier**
as row 1

→ **Presence classifiers**
as row 1

→ **One flow: all slice values**
as row 1, except the vertex:
asinh((reco - true) / 1 cm)
instead of the absolute position;
added back to the true vertex and
clipped to the real range

→ **Result: vertex core fixed, link unchanged**
reco - true vertex median 0.4 / 0.6 cm (exit / contained),
P(> 5 cm) 0.18 / 0.31 (real 0.16 / 0.34; shipped 4-6 cm).
But CC Npi at n_pfp = 3, longest end in FV 0.502 (0.439).
Purity 0.397, eff 0.193 = the shipped chain.
The rest of the link is in dirt neutrinos (3/5 of these
slices): P(exit | reco vertex in FV) real 0.75, gen 0.55;
exiting tracks end on the wall 0.51 real, 0.23 gen
(M2w with wall targets: 0.51). Combined: row 6.

6. Adopted: the shipped slice generator (run m2_official_m2vw, 2026-11-09): row 5 + the wall targets of M2w (longest-track length vs path to wall, cohpi end points: flag + offse

Truth input
as row 5

→ **Topology classifier**
as row 1

→ **Presence classifiers**
as row 1

→ **One flow: all slice values**
as row 5, plus wall targets:
longest length / path to wall - 1
as flag + offset; cohpi muon and
pion end per axis: wall flag + offset;
flagged values snapped to the wall

→ **Result: both zones move, purity 0.400 (eff 0.188)**
held-out split: 0.403 vs previous 0.398 (real 0.425); adopted.
Dirt: P(exit | reco vertex in FV) 0.63 (real 0.75, M2v 0.55);
M2-level link +0.011 (real +0.025); CC Npi at n_pfp = 3,
longest end in FV 0.476 (real 0.439, shipped 0.49).
Gap decomposition: background +194 is the slice
generator (hybrid with real slices: +31); signal -77 is
the PFP generator (cathode side of the track ends:
P(one side | both ends in FV) 0.06 real, 0.22 gen).

Terms

always-present values: vertex, nu_score, bcfm_score, total_pe, longest-track direction / length / PID / chi2 (every slice has them). optional groups: 2nd track (9 values),
two-track block (reco_t, opening angle, range momenta, 12 end points), charge in 1-4 cm spheres. presence flags: has 2nd track, has two-track block, has sphere 1 / 2 / 3 / 4.
counts: n_pfp, n_prong, n_trk, n_proton, n_shower; 121 real combinations + other. the link: in real data a slice whose longest track leaves the TPC tends to have fewer PFPs,
so CC Npi events with an exiting muon fall into n_pfp = 3 (the 2prong bin) more often. Colours: blue truth input, orange learned model, grey deterministic rule, white result.
Models: particle encoder (code m3), slice generator (code m2), PFP generator (code m4).